Table 1: Validity (%), joint MMD, and coverage (%) across five DGPs. Results are means across three seeds. For each CG variant and SEDGE-LGD, one scale per DGP is selected by highest mean validity among $s \in \{0.5, 1, 2\}$; all metrics use that same scale. This is post-hoc test-set selection, not validation-based tuning.

Method	Weighted Discrete			Linear			Cubic			Smoothstep			2D Bridge		
	Valid. ↑	J-MMD ↓	Cover. ↑	Valid. ↑	J-MMD ↓	Cover. ↑	Valid. ↑	J-MMD ↓	Cover. ↑	Valid. ↑	J-MMD ↓	Cover. ↑	Valid. ↑	J-MMD ↓	Cover. ↑
Conditional diffusion (generation direction $Z \to X$)															
CD	13.55	0.1467	40.00	83.22	0.0025	78.52	78.12	0.0062	74.35	85.55	0.0034	70.49	80.65	0.0072	73.08
Compositional CD	75.50	0.1065	84.78	38.54	0.0139	31.25	31.74	0.0368	27.67	21.21	0.0497	16.06	20.93	0.1054	20.04
Extrapolative generation baselines															
Additive mean†	0.00	0.1174	0.00	**100.00**	0.0088	0.55	0.00	0.1711	0.00	0.00	0.0716	0.00	0.00	0.3416	0.00
Additive + residual†	0.18	0.2322	0.65	7.81	0.1407	6.17	0.00	0.2037	0.00	1.35	0.1523	0.86	0.18	0.3266	0.10
Fixed composition + residual†	0.13	0.2353	0.50	0.03	0.2327	0.00	0.02	0.2014	0.00	0.07	0.2548	0.00	0.00	0.3798	0.00
Rep4Ex-inspired + residual†	0.85	0.2714	2.87	3.68	0.2490	3.06	0.23	0.2732	0.02	0.39	0.2621	0.02	0.03	0.3648	0.03
Classifier-guided diffusion (likelihood direction $X_t \to Z$)															
CG: sparse factorized	76.13	0.0530	61.22	24.82	0.0161	7.42	35.74	0.0183	13.85	19.48	0.0383	4.87	36.04	0.0758	30.99
CG: dense factorized	**82.85**	0.0633	56.77	29.61	0.0116	9.83	26.16	0.0194	2.67	17.01	0.0423	1.14	5.70	0.0752	4.64
CG: dense joint	2.03	0.4807	5.93	18.20	0.0111	9.18	34.49	0.0143	13.77	25.72	0.0241	6.35	14.19	0.0394	9.52
SEDGE (ours)															
SEDGE: no weighting	12.00	0.2198	29.95	8.98	0.4006	20.82	8.97	0.3650	17.61	8.92	0.4248	17.12	2.46	0.6294	4.10
SEDGE-LGD	56.80	0.0996	81.08	78.24	0.0104	79.17	71.89	0.0302	50.91	79.57	0.0147	70.05	76.25	0.0516	56.77
SEDGE-LWS	77.55	**0.0225**	**89.97**	91.98	**0.0018**	**87.37**	**91.26**	**0.0024**	**84.29**	**91.55**	**0.0030**	**82.23**	**87.39**	**0.0016**	**78.86**

Selected scales in column order: sparse-factorized CG $(2, 2, 2, 2, 2)$; dense-factorized CG $(2, 2, 2, 2, 2)$; dense-joint CG $(0.5, 2, 2, 2, 2)$; SEDGE-LGD $(1, 1, 2, 1, 2)$. Validity requires both specifications and root support; J-MMD measures joint (X, Z) fidelity; coverage uses the reference-based fixed-radius criterion. Bold marks the best displayed mean per DGP/metric, not statistical significance. †Paper-inspired adaptations, not faithful reproductions; additive mean is a deterministic control. Weighted Discrete has conditional exclusion 2.5%/5%; continuous settings pass the working 1% band-conditional criterion. The expanded LGD sweep was cancelled before execution.